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NPS/001/007 – Technical Specification for Conductors for Overhead Lines

1. Purpose

The purpose of this document is to detail the range of conductors for use on the distribution networks of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/004/049	1.0	Specification for Conductor
NEDL/MS-0/11	Issue 3 Aug 2003	Technical Specification for Conductors for Overhead Lines
E202	Issue 3	Technical Specification for Conductors for Overhead Lines
NPS/001/007	1.0	Technical Specification for Conductors for Overhead Lines
NPS/001/007	1.0	Technical Specification for Conductors for Overhead Lines

2. Scope

This document applies to all Aluminium and Copper based overhead line conductors. The technical particulars are detailed in national, international and industry standards that are listed within the requirements for each type.

3. Technical Requirements

3.1 Condition of Operation

In accordance with this specification, conductors are required to withstand being exposed to the relevant range of climatic conditions. The ambient design temperature range shall be -20° C and 40° C with a maximum conductor operating temperature of 75° C.

3.2 Material Requirements

3.2.1 Stranded all Aluminium Alloy Conductors

The conductors shall be manufactured in accordance with BS EN 50182:2001 and BS EN 50183: 2000. The conductors shall be greased in accordance with the requirements of Engineering Recommendation L38.

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Aluminium Alloy conductors shall be supplied with the conductors in an inert condition as detailed in BS EN 50182 clause 5.5.7.

Manufacturers shall carry out additional tests to confirm the inertness of the completed conductor. This shall involve running out a test length of approx 20m of conductor on the factory floor and visually confirming that the conductor lies flat on the floor. Conductors that exhibit a pigtail or helix effect are not acceptable. This test shall be applied to one sample of conductor from every production run. This frequency may be increased where a history of this problem exists.

This conductor is the standard conductor for new or refurbished HV overhead lines. The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.2 Stranded Aluminium (Steel Reinforced) ACSR Conductors

The conductors shall be manufactured in accordance with BS 215 Part 2: 1970 and BS EN 50182:2001 with the grade and class of zinc coated steel wire shall be manufactured in accordance with BS EN 50189. Conductors shall be greased in accordance with the requirements of Engineering Recommendation L38. This conductor is used for repairs or diversions of existing EHV overhead line Routes.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.3 Stranded Aluminium Alloy (Steel Reinforced) AACSR Conductor

The conductors shall be manufactured in accordance with BS EN 50182:2001 and BS 215, Pt 2: 1970 with the grade and class of zinc coated steel wire manufactured in accordance with BS EN 50189. Conductors shall be greased in accordance with the requirements of ENA Engineering Recommendation L38. This conductor is used for earth wires on L3 construction tower lines.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.4 Plain Hard Drawn Stranded Aluminium Conductor

The conductors shall be manufactured in accordance with BS 215 Part 1: 1970, BS EN 50182:2001, and BS EN 60889. Conductors shall be greased in accordance with the requirements of Engineering Recommendation L38. The prime use for this conductor is for the repair of existing bare LV network conductors

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.5 Aerial Bundled Conductors

LV Aerial Bundled Conductors (ABC) shall be manufactured and tested in accordance with BS 7870-5.

Four core bundles shall consist of three phase cores and one neutral core. Two core bundles shall consist of one phase core and one neutral core.

Five core bundles that include protective conductors shall be as described for 4 core bundles with the addition of a 25mm smooth (no rib markings) insulated core manufactured as detailed in BS 7870-5 Table 1.

Phase cores and neutral cores shall be of un-greased, compacted, circular, stranded aluminium manufactured in accordance with BS 7870-5. They shall be insulated with extruded cross-linked polyethylene and coloured black, with insulation thickness and core diameter in accordance with BS 7870-5 table 1.

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The cores of aerial bundled conductor shall be identified in accordance with Section 3.3 of ENA TS 43 – 13 and laying up shall be in accordance with Section 3.4 of ENA TS 43 – 13.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.6 Compact Covered Conductors

The Compact Covered Conductors (CC) shall be manufactured in accordance with ENA TS 43-122.

The conductor cores shall be circular, stranded and compacted with a longitudinal water barrier. They shall utilise AlMgSi wires according to BS EN 50183 type Al 3 as detailed in ENA TS 43-122 Table 1B.

The conductors shall be designed for operation at up to and including 33KV with a nominal covering of 2.3mm. The prime use for this conductor is for new routes or diversions on HV routes passing through high risk or high tree density areas.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.7 Plain Stranded Hard Drawn Copper Conductors

Bare stranded hard drawn copper conductors shall be manufactured in accordance with the requirements of BS 7884 and shall not be greased. The prime use for this conductor is for repairs or diversions of existing bare HV & LV lines.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.8 Plain Stranded Annealed Copper Conductors

Bare stranded annealed copper conductors shall be manufactured in accordance with BS EN 60228: 2005 class 2, table 2 to provide a high conductivity circular plain conductor. The prime use for this conductor is for jumpers to pole mounted HV plant where it is required to be supported by pilot pin insulators. Where pilot pin insulators are not required, its use shall be substituted by materials specified to clause 3.2.10 of this document.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.9 Hard Drawn, PVC Covered Copper Conductors

The conductor shall be un-greased hard drawn copper manufactured in accordance with BS 7884 with a black PVC covering manufactured in accordance with BS 6485: Table 1, Type 8. The prime uses for this conductor are for LV insulated open wire network conductors.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.10 Single Core Flexible Copper Braid, PVC Covered (Black)

The conductor shall be single core copper flexible braid manufactured generally in accordance with BS EN 60228:2005 class 6 and shall be provided with black PVC covering of in accordance with BS 6004:2000 Table 4b.

The conductor shall be UV resistant with an operating range -40/+105 capable of operating at 750 volts.

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The prime use for this conductor is for unsupported jumpers on HV Pole mounted equipment.

The range of conductors and stranding configurations are detailed in appendix 1.

3.2.11 Single Core Annealed Plain Copper Wire, PVC Insulated

This conductor shall be single core, circular plain annealed copper wire manufactured in accordance with BS EN 60228: 2005 class 2, table 2. It shall be covered with a layer of brown PVC type TI 1 Insulation and provided with a black PVC over-sheath in accordance with BS 6004:2000 table 4A. The conductor shall be designed for operation at 300/500V. The prime use for this conductor is for pole mounted transformer tails to LV fuses.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.12 Single Core stranded copper conductor, PVC Insulated and sheathed

This conductor shall be single core stranded copper conductor to BS EN 60228:2005 class 2 with brown PVC type TI 1 Insulation, covered with a Grey PVC type 6 sheath manufactured in accordance with BS 6004:2000 table 7. The insulation shall be designed for operation at 300/500 V. The prime use for this conductor is for meter tails.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.13 Single Core Annealed Copper Wire, PVC Insulated - Green

This conductor shall be single core, plain stranded annealed copper wire (H07V-R) manufactured in accordance with BS EN 60228:2005 class 2, table 2. It shall be covered with a layer of Green PVC type TI 1 Insulation manufactured in accordance with BS 6004:2000 table 4a. The insulation shall be embossed with the wording "electrical cable" as defined in BS 6346 but with no voltage designation. The insulation thickness shall be enhanced to a 2.0mm thickness with a rated voltage 450/750V. The prime use for this conductor is for Overhead line HV earth leads and bonds.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.14 Single Core Annealed Copper Wire, PVC Insulated - Black

This conductor shall be single core, plain stranded annealed copper wire (H07V-R) manufactured in accordance with BS EN 60228:2005 class 2, table 2. It shall be covered with a layer of Black PVC type TI 1 Insulation manufactured in accordance with BS 6004:2000 table 4a. The insulation shall be embossed with the wording "electrical cable" as defined in BS 6346 but with no voltage designation. The insulation thickness shall be enhanced to a 2.0mm thickness with a rated voltage 450/750V. The prime use for this conductor is for transformer neutral earth leads.

The range of conductors and stranding configurations are detailed in Appendix 1.

3.2.15 Single Core Stranded Copper Wire, PVC Insulated

Single core stranded copper conductor with PVC insulation to 450/750V shall be manufactured in accordance with BS 6004:2000 table 4a

The range of conductors and stranding configurations are detailed in Appendix 1.

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3.3 Greasing Requirements

3.3.1 General

Overhead line conductor greasing shall comply with the requirements of ENA Engineering Recommendation L38.

PVC sheathed copper conductors and LV aerial bundled conductors shall not be greased.

Where required, conductors shall be greased using one of the following materials:

- (a) Ensis 358 or CC manufactured by Shell
- (b) Rustilo 431 manufactured by Burmah-Castrol
- (c) Rustilo 450 manufactured by Burmah-Castrol (cold application)
- (d) Metalube OCG1000 or OCG4500 (Cold Application)
- (e) Metalube OCG3000

The company has a requirement for cold applied grease, where alternatives are offered, this must be noted as a deviation from the specification.

Appendix 1 details each conductor type and the category of grease that shall be applied.

3.3.2 Category of Grease

Category 4 - Fully greased and wiped

Conductors shall not be greased overall and excessive external grease is unacceptable. Conductors must be clean to handle with surplus grease wiped off to produce a clean conductor surface.

Category 3 - Internally fully greased

Conductors shall be greased in accordance with Category 3 of Engineering Recommendation L38

Conductors shall have their inner wires coated with grease and no grease shall be visible on the outer surface of the conductor.

3.4 Testing

Conductors shall be tested in accordance with this specification and the subsequent standards relevant to each particular conductor type. The Self Certification Conformance Declaration (Appendix 3) shall be completed for each conductor.

3.5 Drum and Packaging Requirements

Drum types are detailed for each conductor in Appendix A with specific types and maximum dimensions and weights detailed in Appendix 2.

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4. References

4.1 External Documentation

Reference	Title
BS 7884: 1997	Copper and copper cadmium stranded conductors for overhead electric traction and power transmission systems
BS 215 Part 2: 1970	Aluminium stranded conductors for overhead power transmission
BS EN 50182: 2001	Conductors for overhead lines - Round wire concentric lay stranded conductors.
BS EN 50183: 2000	Conductors for overhead lines – Aluminium-magnesium-silicon alloy wires.
BS EN 60889: 1997	Hard Drawn aluminium wire for overhead line conductors
BS EN 60228:2005	Conductors of insulated cables
BS EN 50189: 2000	Conductors for overhead lines. Zinc coated steel wires
BS 6004: 2000	Electric cables - PVC insulated, non-armoured cable for voltages up to 450/750 Volt for electrical power, lighting and internal wiring
BS 6485: 1999	Specification for PVC covered conductors for overhead power lines
BS 7870 – 5: 1999	LV and MV polymeric insulated cables for use by distribution and generation utilities – Insulated aerial bundled conductors (ABC) of rated voltage 0.6/1 kV for overhead distribution
ENA TS 43-14	Aerial Bundled Conductor. Insulated with Cross-Linked Polyethylene for Low Voltage Overhead Distribution
ENA TS 43-122	XLPE Cover-conductors for overhead lines (greater than 1kV, up to and including 33kV Overhead Lines)
ENA ER L 38	Recommendations for the Protective greasing of PVC-covered copper and bare and PVC-covered aluminium based overhead line conductors

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	02/0/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	05/10/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	02/10/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	26/10/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	06/10/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	06/10/2009

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Appendix 1 – Technical Schedule

Aluminium Alloy (AAAC) to BS EN 50183:2000 (Section 3.2.1)

Cond. (Sq mm)	Stranding Details	BS EN 50182 Description	Conductor Greasing ER L 38	Commodity Code	Std Drum type & Length (m)
50mm	7/3.35mm Hazel	(60-AL3)	Cat 4	226233	RD 1500 or OAR
100mm	7/4.65mm Oak	(119-AL3)	Cat 4	226248	
175mm	19/3.76mm Elm	(211-AL3)	Cat 3	226252	RD 1000 or OAR
200mm	37/2.87mm Popular	(239-AL3)	Cat 3	020859	RD - OAR
300mm	37/3.53mm Upas	(362-AL3)	Cat 3	226253	
500mm	16/3.5mm Rubus	(587-AL3)	Cat 3	226254	

Stranded Aluminium Steel Reinforced (ACSR) to BS 215 – Part 2 (Section 3.2.2)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing ER L 38	Commodity Code	Std Drum type & Length (m)
70mm	12+7 /2.79mm Horse	73-AL1/43- ST1A	Cat 3	226515	RD 1000 or OAR
100mm	6/ 4.72+ 7 /1.57mm Dog	105-AL1/14- ST1A	Cat 3	226534	RD 1000 or OAR
150mm	18+1 /3.35mm Dingo	159-AL1/9- ST1A	Cat 3	226553	RD 1000 or OAR

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150mm	30+7/2.59mm Wolf	158-AL1/37- ST1A	Cat 3	226549	RD 1000 or OAR
175mm	30+7 /2.79mm Lynx	183-AL1/43- ST1A	Cat 3	226568	RD 1000 or OAR
175mm	18+1/3.61mm Caracal	184-AL1/10- ST1A	Cat 3	220627	RD 1000 or OAR

Aluminium Alloy Conductor Steel Reinforced (AACSR) to BS 215, Pt 2 (Section 3.2.3)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing ER L 38	Commodity Code	Std Drum type & Length (m)
160mm	30+7/2.79mm Keziah	183-AL5/43- ST1A	Cat 3	221031	RD OAR

Hard Drawn Aluminium Stranded Conductor to BS 215 – Part 1 (Section 3.2.4)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing	Commodity Code	Std Drum type & Length (m)
50mm	7/ 3.10mm Ant	53-AL1	Cat 4	226337	RD 1000 or OAR
100mm	7/4.39mm Wasp	106-AL1	Cat 4	226360	RD 1000 or OAR

LV Aerial Bundled Conductors (ABC) to BS 7870-5 (Section 3.2.5)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing	Commodity Code	Std Drum type & Length (m)
4 core 35mm	N/A	N/A	N/A	022855	RD 500
4 core 70mm	N/A	N/A	N/A	226031	RD 500
4 core 120mm	N/A	N/A	N/A	226021	RD 500
4 Core 120mm + 25mm Protective cond	N/A	N/A	N/A	226030	RD 500

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Compacted Covered Conductor (CCC) Aluminium Alloy EATS 43-122 (Section 3.2.6)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing	Commodity Code	Std Drum type & Length (m)
50mm	7 strand compacted	N/A	N/A	221033	RD 1000 or OAR
120mm	19 strand compacted	N/A	N/A	221070	
185mm	34 (min) strand compacted	N/A	N/A	221072	

Plain Stranded Hard Drawn Copper Conductors to BS 7884 - Copper conductor shall be high conductivity bare stranded. (Section 3.2.7)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing	Commodity Code	Std Drum type & Length (m)
16	3/2.65mm	N/A	N/A	026534	RD - 500
32	3/3.75mm	N/A	N/A	225264	RD - 500
70	7/3.55mm	N/A	N/A	225298	RD - 500
100	7/4.30mm	N/A	N/A	225300	RD - 500
125	19/2.90mm	N/A	N/A	225315	RD - 500

Plain Stranded Annealed Copper Conductors to BS EN 60228: 2005 class 2, table 2 high conductivity circular plain. (Section 3.2.8)

Conduct or (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Grease	Commodity Code	Std Drum type & Length (m)
32	7/2.46mm	N/A	N/A	222177	C - 100
70	19/2.14mm	N/A	N/A	225584	C - 100

Hard Drawn Copper Conductors (based on BS 7884) PVC Covered to BS 6485: Table 1, Type 8 Black (Section 3.2.9)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Conductor Greasing	Commodity Code	Std Drum type & Length (m)
16	7/1.7mm	N/A	N/A	224740	R - 250
32	7/2.46mm	N/A	N/A	224755	R - 250
70	7/3.55mm	N/A	N/A	224774	R - 250

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Single Core Copper flexible braid PVC Covered (Black) to BS 6004 Table 4b, UV resistant, –15 to 105 ° C operating range rated at 450/750 Volt. (Section 3.2.10)

Cond. (Sq mm)	Stranding Details	BSEN50182 Description	Insulation Colour	Commodity Code	Std Drum type & Length (m)
35mm	12 bunches of 165/0.15mm	N/A	Black	224807	R 500
70mm	13 bunches of 307/0.15mm	N/A	Black	224808	
120mm	12 bunches of 566/0.15mm	N/A	Black	224809	

Single core, annealed copper to BS EN 60228: 2005 class 2, table 2. PVC insulated and sheathed to BS 6004 300/500V. (Section 3.2.11)

Cond. (Sq mm)	Stranding Details	Conductor Greasing ER L 38	Insulation Colour	Commodity Code	Std Drum type & Length (m)
35mm	19/1.53mm	N/A	Brown Ins, Black Sheath	028563	C - 100
95mm	37/1.78mm	N/A	Brown Ins, Black Sheath	028621	C - 100
120mm	37/2.03mm	N/A	Brown Ins, Black Sheath	028647	C – 100
150mm	37/2.25mm	N/A	Brown Ins, Black Sheath	028648	C – 100

Single core stranded copper cable to BS EN 60228 and pvc insulated and sheathed, 300/500 V to BS 6004 table 7, Ref: (Section 3.2.12)

Cond. (Sq mm)	Stranding Details	Conductor Greasing ER L 38	Insulation colour	Commodity Code	Std Drum type & Length (m)
16mm	7/1.70mm	N/A	Brown/Grey	220186	C - 100
16mm	7/1.70mm	N/A	Blue/Grey	220190	C - 100
25mm	7/2.14mm	N/A	Brown/Grey	220203	C - 100
25mm	7/2.14mm	N/A	Blue/Grey	220218	C - 100

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Annealed high conductivity copper circular conductors to BS EN 60228: 2005 class 2, table 2, PVC insulated to BS 6004 colour Green embossed 'electrical cable' as BS 6346 but with no voltage designation, with enhanced nominal radial thickness of PVC covering to 2.0mm. (Section 3.2.13)

Cond. (Sq mm)	Stranding Details	Conductor Greasing ER L 38	Insulation colour	Commodity Code	Std Drum type & Length (m)
32mm (Green)	7/2.46mm	N/A	Green	220147	R - 250
70mm (Green)	7/3.55mm	N/A	Green	220146	R - 250

Annealed high conductivity copper circular conductors to BS EN 60228: 2005 class 2, table 2, PVC insulated to BS 6004 colour Black embossed 'electrical cable' as BS 6346 but with no voltage designation, with enhanced nominal radial thickness of PVC covering to 2.0mm. (Section 3.2.14)

Cond. (Sq mm)	Stranding Details	Conductor Greasing ER L 38	Insulation colour	Commodity Code	Std Drum type & Length (m)
32mm (Black)	7/2.46mm	N/A	Black	224806	R - 250

Single core stranded copper cable to BS EN 60228: 2005 class 2, table 2., PVC insulated only, 450/750 V to BS 6004 table 4a (Section 3.2.15)

Cond. (Sq mm)	Stranding Details	Conductor Greasing ER L 38	Insulation Colour	Commodity Code	Std Drum type & Length (m)
6.0mm	7/1.04mm	N/A	Green/yellow	220148	C – 100
16.0mm	7/1.70mm	N/A	Green/yellow	027672	C – 100
25.0mm	7/2.14mm	N/A	Green/yellow	027730	C - 50
35.0mm	19/1.53mm	N/A	Green/yellow	027771	C – 50
70.0mm	19/2.14mm	N/A	Green/yellow	027854	C – 50
120.0mm	37/2.03mm	N/A	Green/yellow	027904	C – 50

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Appendix 2 – Logistical Requirements

Drum and Packaging Requirements

All overhead line conductors shall be supplied on returnable drums (RD), on reels (R) or in coils (C) as specified in Appendix 1. Standard drum lengths are detailed in Appendix 1 together with a requirement for section lengths to be ordered as required (OAR).

Returnable Drums (RD)

Standard Drum Dimensions – Maximum flange diameter 1200mm, maximum drum width of 1000mm (this dimension shall include any bolt heads or studs located on the side of the drum) and a minimum barrel diameter of 600mm except when the drum is used to carry 100mm² AAAC. Then the minimum diameter shall be 800mm. The spindle hole shall have a diameter between 80 – 100mm. The gross weight shall not exceed 670kg.

ABC Drum Dimensions - Maximum flange diameter of 1800mm, maximum drum width of 1100mm (this dimension shall include any bolt heads or studs located on the side of the drum) and a minimum barrel diameter of 30 x the diameter of the conductor or a minimum of 800mm. The spindle hole shall have a diameter between 85 – 100mm. The gross weight shall not exceed 1300kg.

For all drum types the two outer flanges shall be drilled with two 25mm diameter holes opposite to each other and 275mm from the central spindle hole.

All drums shall be designed to take a round spindle and be lagged to protect the conductor, whilst on the drums, from the risk of damage during transportation and handling on site.

The inner end of the conductor projecting from the drum shall be secured and protected to avoid damage with the direction of rolling being indicated on the outer flange of drum.

All returnable drums shall be labelled on one flange and the label must state:

- (a) Manufacturer's name
- (b) Conductor type
- (c) Conductor cross section and stranding
- (d) Conductor length in metres (either standard or OAR)
- (e) Drum net weight excluding lagging
- (f) Drum gross weight
- (g) The Company order number and date
- (h) The Company commodity code

Labelling shall be by any method that fulfils all of the following criteria:

- (a) is not affected by rain or other adverse weather
- (b) is not affected by ultra violet light
- (c) is, and remains, legible

However the use of cards or papers whether or not enclosed is **NOT ACCEPTABLE**.

Drums shall be lagged using weatherproof wood fibreboard (an example of this is Nolco-flex) providing suitable protection to the conductor and secured with a circumferential banding system. Drums with battens nailed to the flanges **ARE NOT ACCEPTABLE**.

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Drums shall not be protected with any secondary protection such as water resistant paper.

Reels (R)

Reels shall have a maximum height of 750mm, maximum width of 400mm and a minimum diameter of 250mm. The spindle hole shall have a diameter between 80 – 100mm. All other requirements shall be in accordance with those specified for returnable drums.

Coils (C)

Where conductors are supplied in coils the coil length shall be 50 or 100m secured by suitable cable ties. The coil diameter shall ensure the cable is not damaged and suitable for transportation.

Covered Conductor

All insulated or covered conductor ends shall be suitably sealed to prevent the ingress of moisture

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Appendix 3 – Self Certification Conformance Declaration

Conductors for overhead lines shall comply with the latest issues of the relevant international and British Standards.

The relevant BS, BS EN and ENA Technical Specifications are quoted to amplify and/or clarify the requirements of those Standards.
This check sheet identifies the particular clauses of the aforementioned Standards relevant to overhead line conductors.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes for each conductor.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Manufacturer:

Product Reference:

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Name:

Signature:

Date:

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All aluminium alloy conductors (AAAC) – Section 3.2.1

Stranded aluminium steel reinforced (ACSR) – Section 3.2.2

Stranded aluminium alloy steel reinforced (AACSR) – Section 3.2.3

Plain hard drawn stranded aluminium – 3.2.4

BS EN 50183: 2000			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
4	Materials		
5 - Table 1	Properties of heat treated wires		
6	Diameter and tolerance on diameter		
7	Wire surface		
9	Joints		
10	Sampling		
11	Tests		
BS 215-1: 1970			
1.3	Standards for hard-drawn aluminium wire		
2	Materials		
3	Dimensions and construction		
4	Tests		
BS EN 50182: 2001			
5	Requirements for stranded conductor		
6	Tests		
7	Packaging and marking		

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Annex F – Tables 39-43	Conductor characteristics		
ER L38			
1	General		
2	Properties		
3	Quality Control		
4	Categories (where applicable)		
Table 1 and 2	Conductor Greased Weights		

Aerial Bundled Conductor - Section 3.2.5

BS 7870 – 5: 1999			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
3	Voltage designation		
5	Insulation thickness and core diameter requirements – Table 1		
7	Lay Length – Table 2		
10	Routine tests		
11	Sample tests		
12	Type tests		
13	Sealing and drumming		
ENA TS 43-13			
3	Construction requirements		
4	Test Requirements		

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Compact Covered Conductor - Section 3.2.6

ENA TS 43-122 Issue 1: March 2002			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
4.2	Conductor construction		
4.2.2	Voltage designation – Designed for operation up to and including 33kV.		
Table 1B	Conductor covering, thickness and technical parameters – Table 1		
Appendix A1	Current leakage Test		
Appendix A2	Longitudinal Water Tightness Test		
Appendix A3	Slippage test		
Appendix A4	Stripping Tests		

Compact Covered Conductor - Section 3.2.6

BS EN 60811-1-1, BS EN 60811-2-1, BS EN 60811-3-1 and BS 2782 part 5			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
Clause 9	Mechanical Properties without ageing (a) Tensile Strength (b) Elongation at break		
Clause 9.2	Water absorption of insulation 14 days at 85 °C		
Clause 11	Carbon Black Content		
Clause 10	Shrinkage test 1 hour at 130 °C		

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Compact Covered Conductor - Section 3.2.6

BS EN 60811-1-1, BS EN 60811-2-1, BS EN 60811-3-1 and BS 2782 part 5			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
Clause 9 135 °C for 7 days	Compatibility Test Samples of complete covered conductor shall be aged in accordance with BS EN 60811-1-2 clause 8.1.4 (a) Tensile Strength (b) Elongation at break (c) Max variation a. Tensile Strength b. Elongation at break		
Clause 8 135 °C for 7 days	Mechanical Properties after ageing (a) Tensile Strength (b) Elongation at break		
BS EN 60811-3-1 Clause 8.1	Pressure test at high temperature Temperature test Max depth of indentation		
BS EN 60811-2-1	Hot Set tests (15mins at 0.2N/mm ²) Max elongation under load Maximum elongation after cooling		
BS 2782 part 5 Method 540F	Mechanical properties after UV exposure (UV stabilisers other than carbon black) Max variation from unexposed conductors (a) Tensile strength (b) Elongation at break		

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Plain Stranded Hard Drawn Copper Conductors - Section 3.2.7

BS 7884: 1997			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
4.1	Composition requirements		
Table 3	Physical, mechanical and electrical properties		
Table 5	Lay ratios of stranded conductors		
6	Selection of test samples		
8	Marking, labelling and packaging		

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Plain stranded annealed copper conductors – Section 3.2.8

Single core flexible braid, PVC covered black – Section 3.2.12

Single core annealed plain copper wire, PVC insulated and sheathed – Section 3.2.12

Single core stranded copper cable, PVC insulated PVC sheathed – Section 3.2.13

Single core annealed copper wire, PVC insulated – Section 3.2.14

Single core stranded copper wire, PVC insulated – Section 3.2.15

BS 6360: 1991 - Conductor			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
4	Class 2, 5 and 6		
6.2	Stranded circular non-compacted conductor		
7	Flexible conductors		
9.3	Tests for hard-drawn copper conductors		
Tables 2, 3 and 5	Stranding details		
Table 12	Additional data for annealed copper conductors		
Table 14	Additional data for flexible copper conductors		
BS 6004: 2000 - Insulation			
5	General requirements		
6	Constructional requirements		
7	Electrical tests		
8	Non-electrical tests		

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Hard drawn PVC covered copper conductors – Section 3.2.10

BS 7884: 1997			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
4.1	Composition		
4.2.3	Stranded conductors		
4.3	Mechanical properties		
4.4	Electrical resistance		
4.5	Manufacture		
6	Selection of test samples		
BS 6485 Type 16 for Green Insulation			
BS 6485 Type 8 for Black Insulation			